

Miao Xu

EDUCATION

Lund University <i>Master in Machine Learning, Systems and Control (Sweden) (Full Scholarship)</i> Courses: Learning-based Control, Computer Vision, Image Analysis, Introduction to Artificial Neural Networks and Deep Learning, Monte Carlo and Empirical Methods for Stochastic Inference, Modelling and Learning from Data.	2025 – Present
ITMO University & Hangzhou Dianzi University (dual degree) <i>Bachelor in Control in Technical Systems with Honours (2 %) (Russia) & Bachelor in Automation (China)</i> - GPA: 5.0/5.0 (Russia), 4.69/5.0 (China) (Rank: 2/93) IELTS: 7.0 (L: 8.5, R: 8.0, W: 6.0, S: 6.0) - Courses: Higher Mathematics (98), Theoretical Mechanics (100), Identification Theory (100), Automatic Control Theory (98), Machine Learning & Deep Learning (96), Computer Vision (100), Computational Mathematics (99), Adaptive and Robust control (100), Nonlinear Control Systems (100), Optimal Control (100), Algorithms and Data Structures (97). - Thesis: Research on Control Algorithm of Quadcopter (Outstanding Undergraduate Thesis)	2019 – 2023

PUBLICATIONS

[1] Ren, Q., **Xu, M.**, Zhang, M., Chen, N., Lai, M., Xu, C., Gao, F. and Cao, Y., 2025. COVER: cross-vehicle transition framework for quadrotor control in air-ground cooperation. *Autonomous Robots*, 49(3), p.23.

RESEARCH EXPERIENCE

FAST Lab, Zhejiang University Advisor: Yanjun Cao <i>Research Assistant</i> Air-Ground Cooperative (AGC) Framework for Inter-Vehicle Transfer Tasks - Developed a multiple non-inertial frames based AGC framework and Proposed a differential model for UAVs transferring between UGVs, eliminating the need for high-frequency trajectory replanning and GPS in traditional AGC frameworks. - Designed a phase-adaptive Model Predictive Controller (MPC) with a novel position reference scheme. - Paper [1] has been published in <i>Autonomous Robots</i>. Robust Landing of Unmanned Aerial Vehicles on Dynamic Platforms - Developing a Robocentric frame based differential model, which using only relative visual information between robots. - Developing an L1-Adaptive MPC method that aims to estimate unobservable variables (e.g., acceleration and angular acceleration of the target platform) in the equation of state, potentially enhancing the adaptability of the control system.	Zhejiang, China 2023 – 2025
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PROJECT EXPERIENCE

Detect empty drums on the exterior walls of buildings using a quadrotor <i>Intelligent sound detection project</i> Advisor: Chao Xu , Yanjun Cao Designed an innovative quadrotor capable of flying closely along walls, moving smoothly, and tapping surfaces for sound signal collection, offering an effective solution for detecting hollow drums in building exteriors.	Zhejiang, China 2023 – 2024
Research on Control Algorithm of Quadcopter <i>Undergraduate Thesis</i> Advisor: Ke Yin, Sergei Shavetov - Designed a Cascade Active Disturbance Rejection Controller for a quadcopter, ensuring precise tracking under disturbances. - Validated robustness via numerical simulations and effectiveness through hardware-in-the-loop tests. - Received 'Outstanding Undergraduate Thesis' award and selected for exhibition.	Zhejiang, China 2022 – 2023
Research and implementation of a portable rescue fixed-wing UAV <i>Zhejiang Province 'New Seedling' Project</i> Advisor: Yi Luo - Designed a portable rescue UAV for autonomous patrolling and detecting disaster areas with communication loss. - Developed a visual system for post-disaster reconstruction, semantic segmentation, and object detection using machine learning and deep learning techniques.	Zhejiang, China 2021 – 2021

AWARDS

Excellent Graduate of Zhejiang Province (4 %)	2023
Outstanding Paper Award for "SHENZHEN CUP" Mathematical Modeling Challenge (4 %)	2021
National Second Prize for China Aeromodelling Design Challenge (15 %)	2020 & 2021
Scholarship of Zhejiang Province (5 %)	2020 & 2021 & 2022
The First Prize Scholarship of Hangzhou Dianzi University (5 %)	2020 & 2021 & 2022

SKILLS

Programming: C++ (3+ years), C (5+ years), Python (4+ years), ROS (3+ years), MATLAB (4+ years).
Software & Hardware Development: Linux, ROS, Gazebo, Git, Docker, SolidWorks, AutoCAD.